

Investigation of Grain Counting Inconsistencies in MYflow

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Software: MYflow (Casini et al., 2023)

Introduction

Recent evaluations of MYflow 2.0 utilized both the provided sample datasets and custom-generated composite phase maps for validation. Throughout this testing phase, a noticeable inconsistency emerged between the visual count of distinct grains within the composite phase maps and the figures output by the software. This irregularity directly prompted an in-depth review of the grain-identification procedures applied during composite-model processing.

This document aims to detail the findings from the testing period, the resulting code-level analysis, and the adjustments tested to determine the root cause of the variation. While the primary focus remains on phase-map processing and grain quantification in composite datasets, this report also highlights several usability and stability enhancements discovered during the troubleshooting process. These results are presented cautiously to facilitate continued assessment and collaborative review, rather than to serve as absolute determinations of the software's performance across every potential scenario.

Testing Environment

All testing in this report was done using MYflow 2.0 in MATLAB R2026a on Windows 11. The investigation focused on the composite-model workflow, specifically looking at how phase maps are created and analyzed, as well as how grains are identified, counted, and measured.

To start, we tested the software using the original example dataset that comes with MYflow. The software ran successfully and gave results that matched the expected example. After that, we analyzed additional custom-made phase maps that featured multiple minerals and simple grain shapes we could easily verify by hand. We used these extra datasets to test how the grain-identification process handled controlled conditions and to investigate the mismatched grain counts we had noticed earlier.

Baseline Verification Using Distributed Example

Before investigating the observed grain-count discrepancies, the distributed composite-model example supplied with MYflow was processed using the original implementation without modification. This step was performed to verify that the software executed correctly under the conditions intended by the original workflow.

The example dataset was analysed successfully, and the generated phase map, grain-size distributions, and phase-specific grain counts appeared consistent with the visual characteristics of the input image. No runtime errors, unexpected behaviour, or obvious inconsistencies were observed during this initial test. The software therefore appeared to function as intended for the provided example dataset.

This baseline verification was important because it established that the subsequent observations did not arise from installation issues, MATLAB compatibility problems, or incorrect software configuration. Instead, the discrepancies described in the following sections became apparent only during testing with additional composite phase maps designed to evaluate grain identification under different phase distributions and grain geometries.

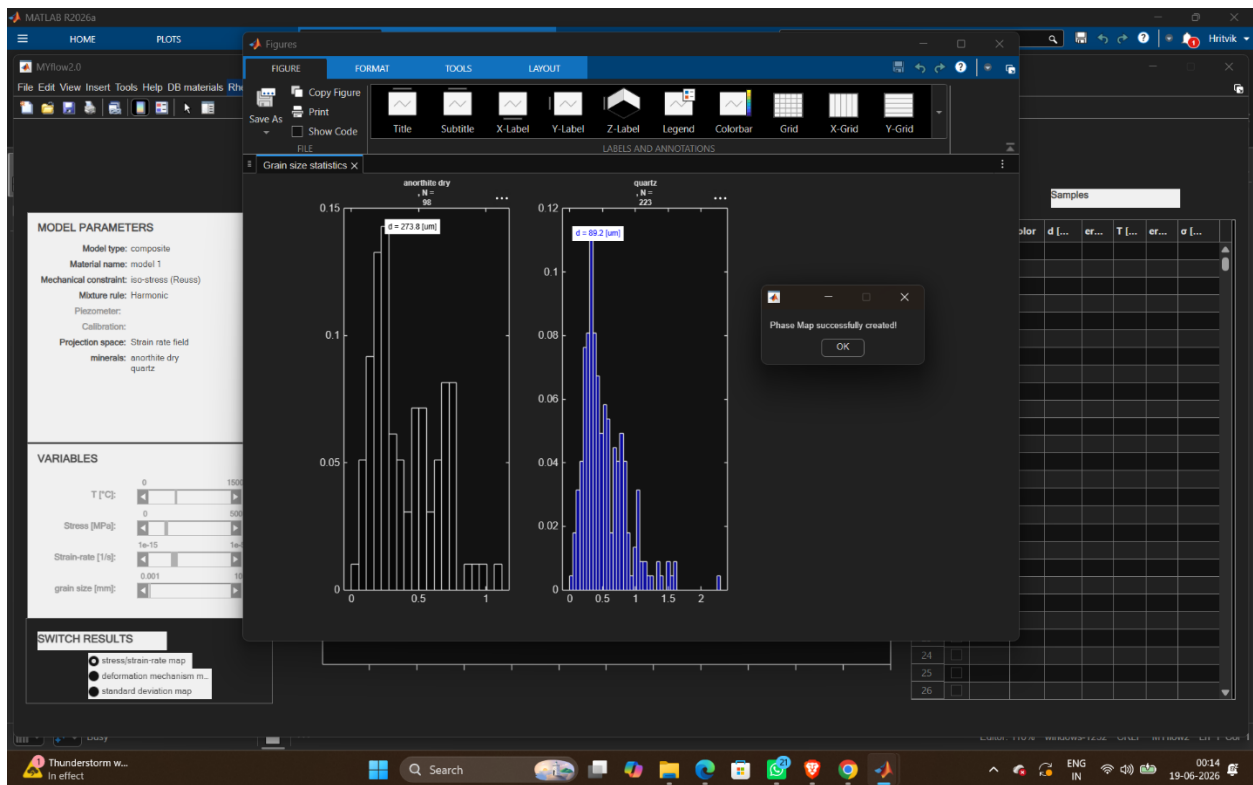


Fig 1: Baseline testing output of Model 1 (provided by the publishers)

Investigation of Composite Grain Count Discrepancies

Observed Behaviour

Following successful verification of the distributed example dataset, additional synthetic composite phase maps were analysed to further evaluate the grain-counting workflow. One such test image contained four mineral phases represented by distinct colours, with grain geometries that could be readily verified through visual inspection.

During analysis of this dataset, an apparent discrepancy was observed between the number of visually distinguishable grains and the grain counts reported by MYflow for one of the mineral phases. In particular, (refer to fig 2b) the dark-blue phase (K-feldspar) contained four clearly separated grains within the analysed region. However, the grain-size statistics generated by the original implementation reported a total of seven grains for this phase.

Because the grains were well separated and visually identifiable, the reported count appeared higher than expected. At this stage, the source of the discrepancy was unknown, and no assumptions were made regarding its cause. The observation nevertheless suggested that some aspect of the composite-model grain-identification workflow might be influencing phase-specific grain counts. This finding motivated a detailed investigation of the image-processing and grain-labelling procedures used during composite-mode analysis.

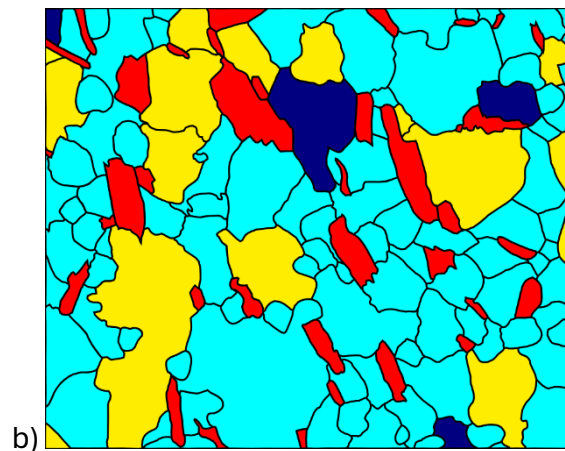
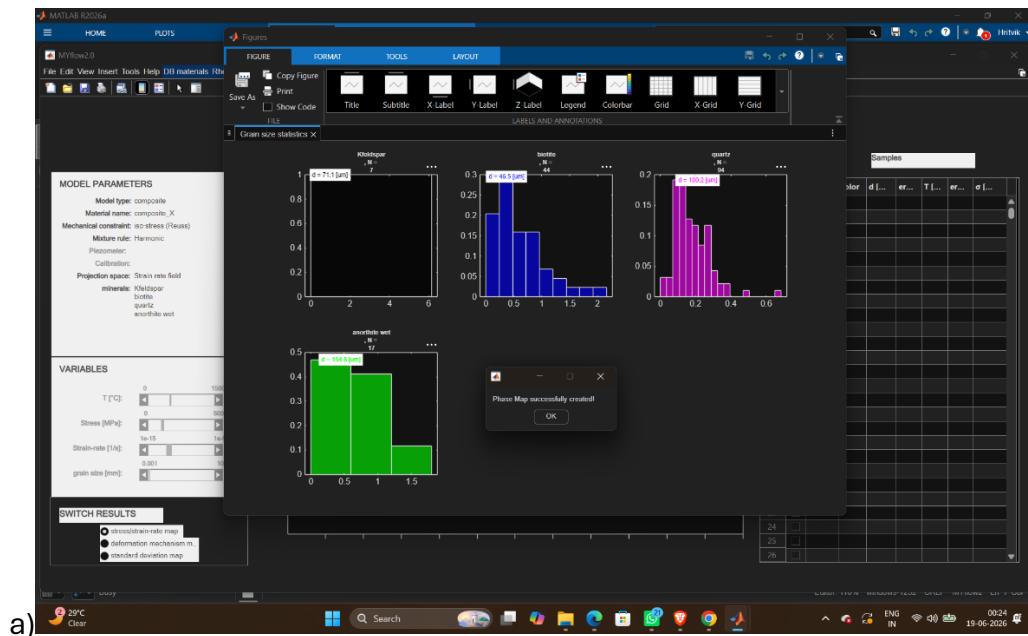


Fig 2: Observation on a different phase map

Investigation

To identify the source of the observed grain-count discrepancy, the composite-mode workflow was examined step-by-step, with particular attention given to phase identification, boundary detection, connected-component labelling, and grain counting. The investigation focused on tracing how the input phase map was transformed into the binary masks used for grain identification.

During this process, it was observed that the original implementation relied on the MATLAB function `rgb2ind(im, N+1)` early in the composite-mode workflow (in file “grainsize.m”). This operation converts the original RGB image into an indexed image containing only $N+1$ colour classes, where N is the number of mineral phases selected by the user. Because this step performs colour quantization, all image colours are reassigned to a limited set of representative colour indices.

Further testing suggested that this quantization step could influence the representation of grain boundaries in composite phase maps. In particular, boundary pixels represented by near-black colours were not always preserved as a distinct class after quantization. Depending on the local colour distribution within the image, some boundary pixels appeared to be reassigned to neighbouring phase colours. As a result, grain boundaries could become fragmented, partially removed, or locally altered prior to connected-component analysis.

Because grain identification in MYflow ultimately depends on the separation of neighbouring grains by boundary pixels, modifications to the boundary network may influence the number and geometry of the connected components identified for a given phase. This behaviour provided a plausible explanation for the grain-count discrepancies observed during testing and motivated a series of targeted code modifications to evaluate the effect of preserving boundary information directly from the original RGB image.

Modifications Implemented

In the tested phase maps, grain boundaries were represented by pixels that were black or near-black, whereas mineral phases were represented by distinctly coloured regions with substantially higher RGB intensity values. To preserve these boundaries during segmentation, the modified implementation identified boundary pixels directly from the original RGB image using an intensity-based threshold applied independently to each colour channel.

A pixel was classified as a boundary pixel when all three RGB channel values satisfied the condition:

$$R < 30, G < 30, \text{ and } B < 30.$$

The threshold value of 30 was selected because it provided a clear separation between the dark boundary pixels and the coloured phase regions present in the tested datasets. Pure black boundary pixels correspond to RGB values of (0,0,0), while anti-aliased boundary

pixels and compression-related intensity variations typically remain within a low-intensity range. The chosen threshold therefore allowed both pure black and slightly brighter boundary pixels to be identified reliably while avoiding misclassification of mineral-phase pixels.

For example, a pixel with RGB values of (12,18,15) would be classified as a boundary pixel, whereas a dark-blue phase pixel such as (0,0,255), a red phase pixel such as (255,0,0), or a green phase pixel such as (0,255,0) would not satisfy the threshold condition and would therefore remain assigned to their respective phases. Similarly, moderately dark coloured pixels containing at least one channel value greater than 30 would not be interpreted as boundaries.

By applying the threshold directly to the original RGB image, the boundary mask could be generated without first reducing the image to a limited number of indexed colour classes. This ensured that the boundary network used for connected-component analysis was derived directly from the original phase map and remained independent of the colour quantization process investigated in the previous section.

Results

Following implementation of the modified boundary-detection workflow, the composite phase maps used during testing were re-analysed and the resulting grain counts were compared with those obtained from the original implementation.

For the synthetic composite phase map described in Section 4.1, visual inspection indicated the presence of four distinct K-feldspar grains within the analysed region. When processed using the original implementation, MYflow reported a grain count of seven for this phase. After applying the modifications described in Section 4.3, the reported grain count was reduced to four, which was consistent with the number of visually distinguishable grains observed in the phase map.

In addition to the numerical change in grain count, examination of the intermediate segmentation outputs suggested that the modified workflow preserved grain boundaries more consistently throughout the analysis. The resulting grain masks appeared to better reflect the geometry of the original phase map, particularly in regions where grains were separated by narrow boundary networks.

A comparison of the phase maps and grain-count outputs generated by the original and modified implementations is provided in Figures 2 and 3. The differences observed between the two outputs were consistent with the hypothesis that alterations to the boundary network can influence connected-component identification and consequently affect phase-specific grain counts.

It should be noted that the present investigation was performed on a limited number of test datasets and was intended primarily as a debugging exercise. While the modified implementation produced results that were more consistent with visual inspection in the

tested examples, additional validation using a broader range of composite phase maps would be required to determine the general applicability of these observations.

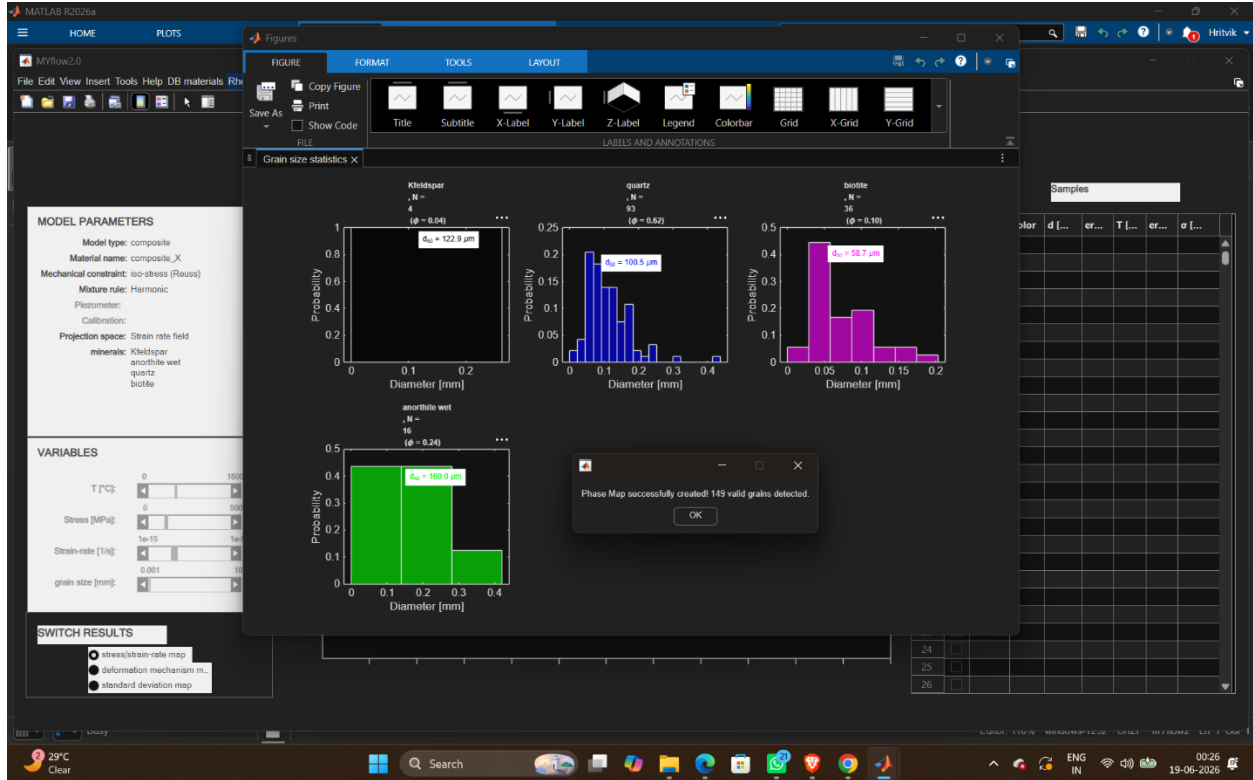


Fig 3: The output of the new code for the same sample

Additional Robustness Improvements

In addition to the modifications related to composite-mode grain counting, several smaller improvements were implemented to enhance the stability and reliability of the grain-size analysis workflow. Although these changes are not expected to significantly affect scientific results under normal operating conditions, they reduce the likelihood of user errors, unexpected behaviour, and software failures in specific edge cases.

Phase Selection Validation

The original implementation relied on user-selected points for phase identification during composite-mode setup. However, no explicit validation was performed to ensure that the selected pixel belonged to a grain interior rather than a grain boundary.

To improve robustness, an additional validation step was introduced to verify the selected pixel location before phase assignment. If a boundary pixel is selected, the user is prompted to choose another location within the grain. This reduces the possibility of incorrect phase identification arising from accidental boundary selection.

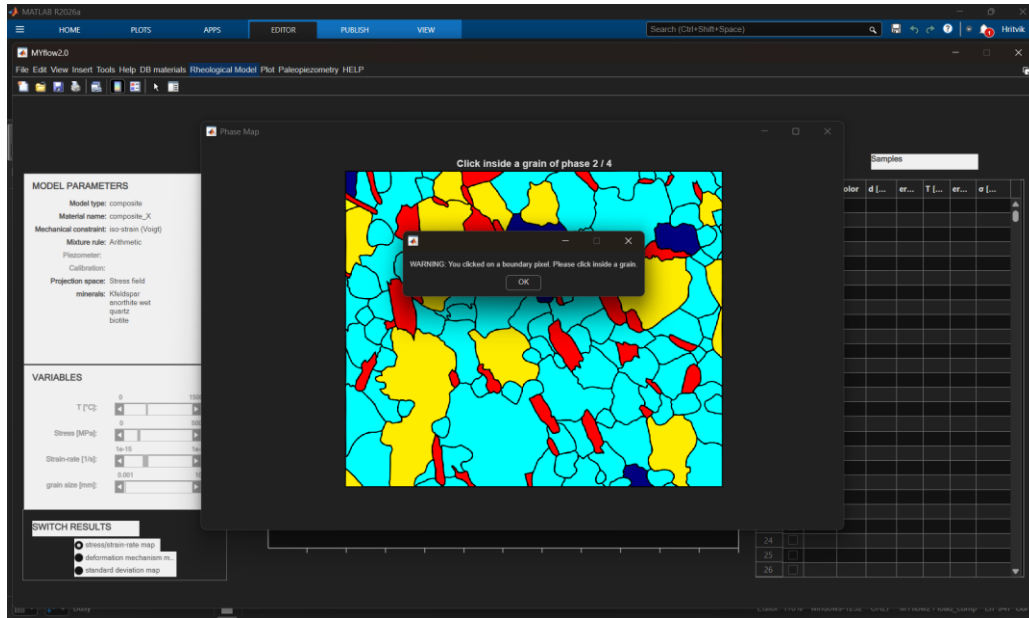


Fig 4: The error message on accidentally clicking the boundary pixel

Duplicate Phase Protection

During composite-mode setup, each mineral phase should be assigned only once. The original implementation used a logical condition to detect duplicate phase assignments, but the behaviour of this check was not always intuitive and could potentially lead to inconsistent phase registration.

The duplicate-detection logic was replaced with a direct comparison against previously assigned phases. If a mineral has already been assigned, the user is notified and prompted to select a different phase. This modification simplifies the logic and improves the reliability of phase registration.

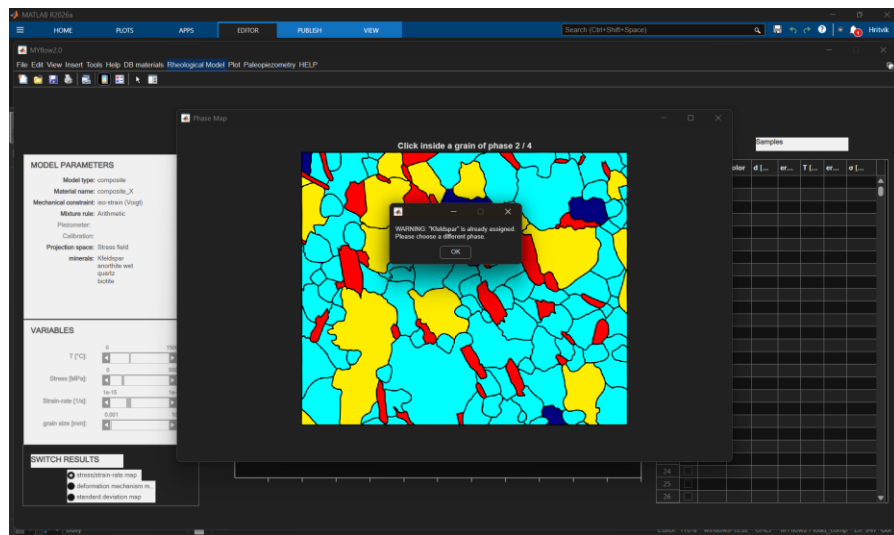


Fig 5: The error message on selecting a different phase for same mineral

Removal of Zero-Label Components

Following phase assignment, some pixels may remain unclassified if they do not correspond to any of the selected mineral phases. These pixels are assigned a label value of zero.

To prevent unclassified regions from contributing to grain statistics, connected components whose dominant label corresponds to zero are excluded from the final grain population. This reduces the likelihood of spurious grains influencing grain-size calculations and phase statistics.

Histogram Bin Edge Case

The original histogram-generation routine determined the number of bins as a function of the number of grains in a given phase. For very small grain populations, this calculation could produce a value of zero, which is not accepted by MATLAB's histogram function and may result in a runtime error.

A lower bound of one bin was introduced to ensure that histogram generation remains valid regardless of sample size. This modification improves software stability when analysing datasets containing only a few grains of a particular phase.

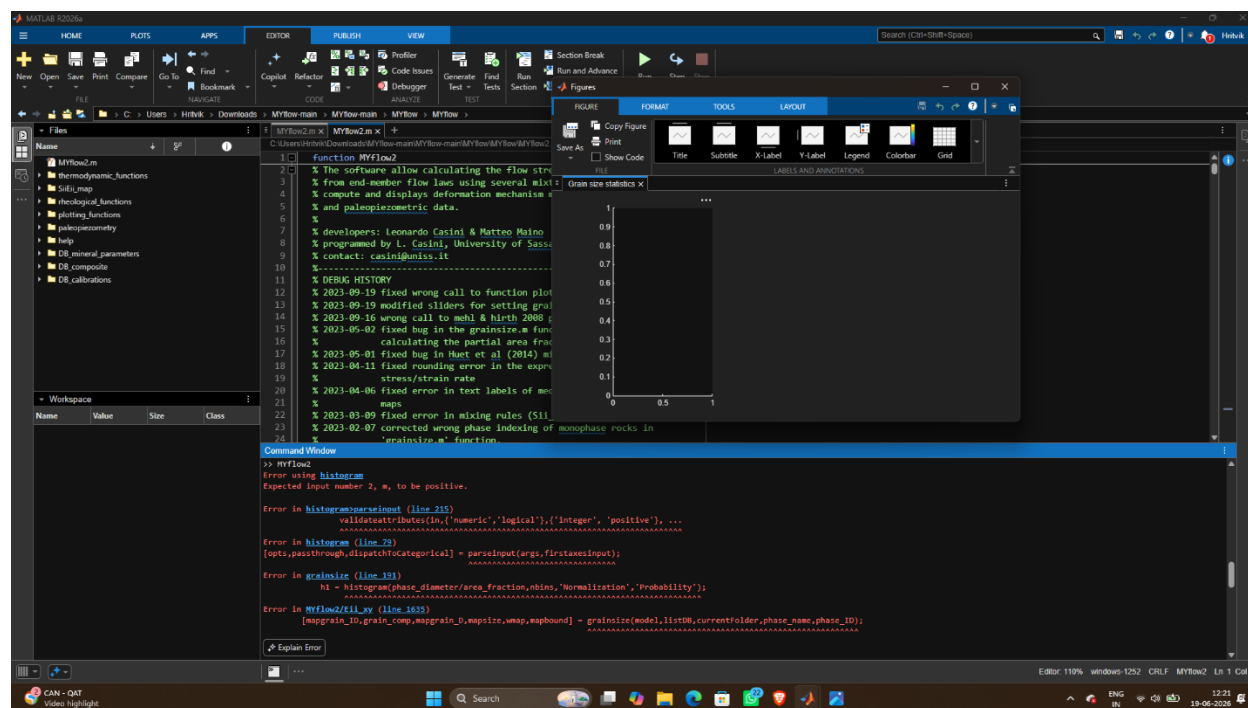


Fig 6: Runtime error generated when the histogram bin calculation returned zero for a phase containing a very small grain population. The modified implementation prevents this condition by enforcing a minimum bin count of one.

Grain-Size Histogram Scaling

The original implementation displayed grain-size distributions using grain diameters scaled by phase area fraction. While this did not affect the underlying grain-size calculations, it altered the visual representation of the diameter distribution and could complicate interpretation of the histogram axis.

The histogram routine was modified to display the measured grain diameters directly. This provides a more intuitive visualization of grain-size distributions while preserving the original statistical calculations.

Conclusion

This investigation began after noticing discrepancies in grain counts during composite-mode analysis in MYflow. Although the software worked perfectly with the default example dataset, testing with additional custom phase maps showed that the reported grain counts did not always match what could be seen visually on the map.

A close look at the workflow suggested that the boundary-extraction step might be affecting how grains are identified. Specifically, reducing the number of colors (color quantization) before finding boundaries seemed to alter the network used to separate the grains. To test this idea, the workflow was changed so that boundary information was taken directly from the original RGB image instead of the color-reduced version.

Using this updated method on the test datasets produced grain counts that matched the visual populations much better. In the main example used in this report, the K-feldspar grain count dropped from seven to four, which exactly matches the manual count from the phase map. The new results also preserved the true shape of the grain boundaries more accurately, especially in areas with very narrow borders.

Alongside the main investigation, several small changes were made to improve the software's stability. These included checks for phase selection, preventing duplicate phases, handling unclassified components, and adding safety checks for generating histograms. While these tweaks are unlikely to change scientific results during normal use, they improve reliability and help prevent user errors in unusual situations.

It should be noted that this study was a debugging exercise using a small number of test datasets. Therefore, these observations should not be seen as a final judgment on how MYflow performs across all possible phase maps. Further testing with a wider variety of datasets is needed to see if these proposed changes apply generally.

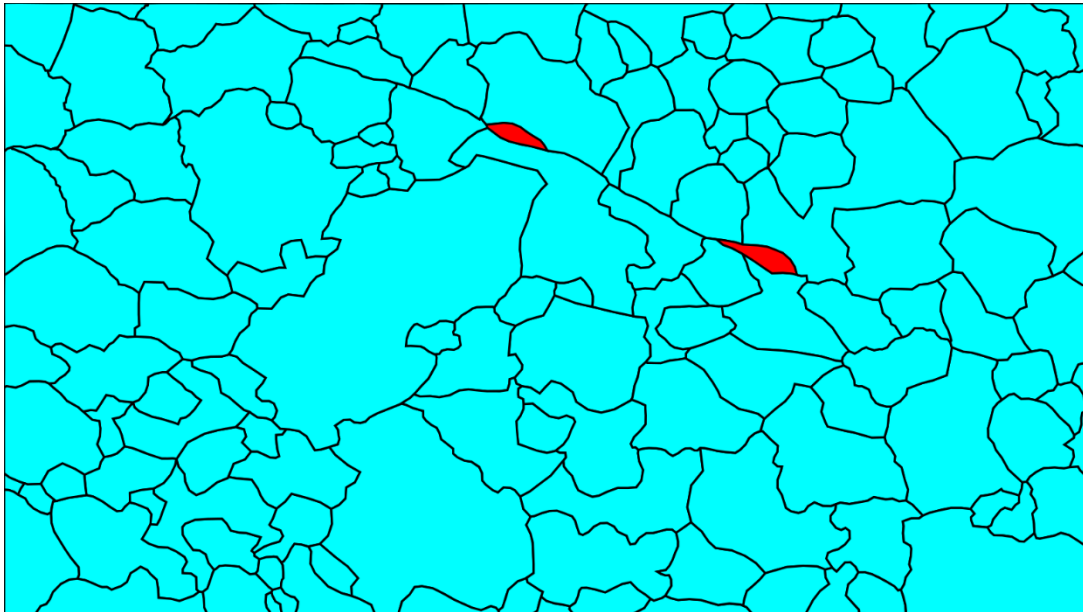
The findings and code updates detailed in this report are shared in the spirit of collaborative software improvement. They are provided for consideration in future development, validation, and reviews of MYflow.

Summary Table:

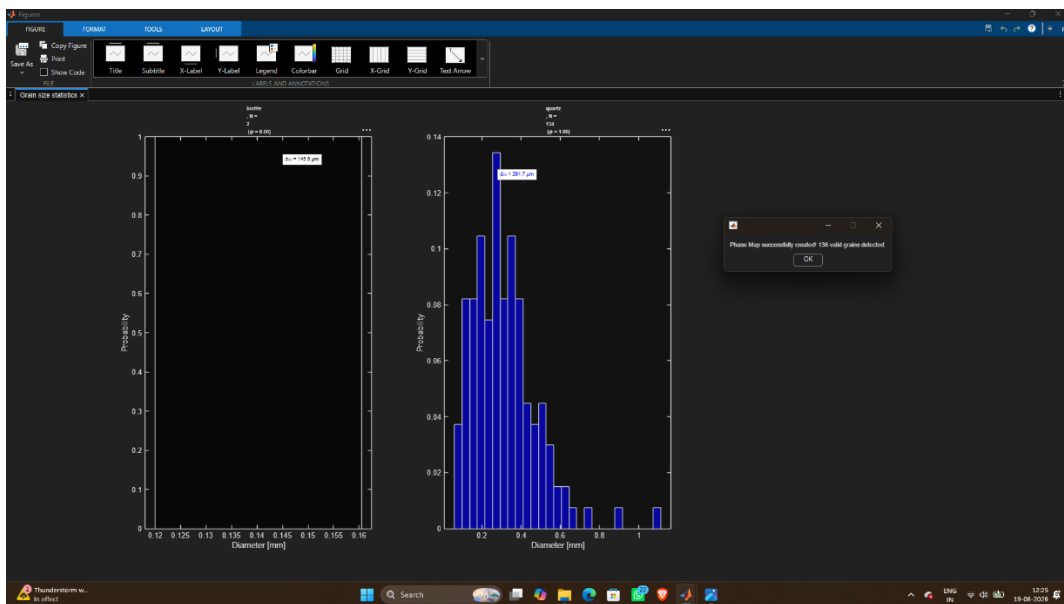
Modification	Original Behaviour	Implemented Change	Expected Impact
Boundary Detection Workflow	Grain boundaries were identified after RGB-to-indexed colour quantization using <code>rgb2ind(im,N+1)</code> .	Boundary pixels were identified directly from the original RGB image using an intensity threshold.	Improved preservation of grain-boundary geometry and reduced sensitivity to colour quantization during grain identification.
Phase Selection Validation	User-selected pixels were accepted without verification of whether they belonged to a grain interior or a boundary.	Boundary-pixel validation was introduced prior to phase assignment.	Reduced likelihood of incorrect phase identification caused by accidental boundary selection.
Duplicate Phase Protection	Duplicate phase assignment was checked using an indirect logical condition.	Replaced with an explicit comparison against previously assigned phases.	Improved reliability and transparency of phase registration during composite-mode setup.
Zero-Label Component Filtering	Unclassified regions could remain present during grain-statistics calculations.	Components dominated by label value zero were excluded from the final grain population.	Reduced contribution of spurious or unclassified regions to grain-size statistics.
Histogram Bin Edge Case	Very small grain populations could produce a histogram bin count of zero, leading to potential runtime errors.	A minimum bin count of one was enforced.	Improved stability of histogram generation for small datasets.
Grain-Size Histogram Scaling	Histograms were displayed using diameters scaled by phase area fraction.	Histograms were generated using the measured grain diameters directly.	Improved interpretability of grain-size distribution plots while preserving underlying calculations.
Input Validation and Error Handling	Limited validation of user inputs and analysis parameters.	Additional validation checks and safeguards were introduced.	Improved software robustness and reduced likelihood of user-facing errors.

Table 1: Summary of the principal modifications investigated and implemented during the debugging and validation process.

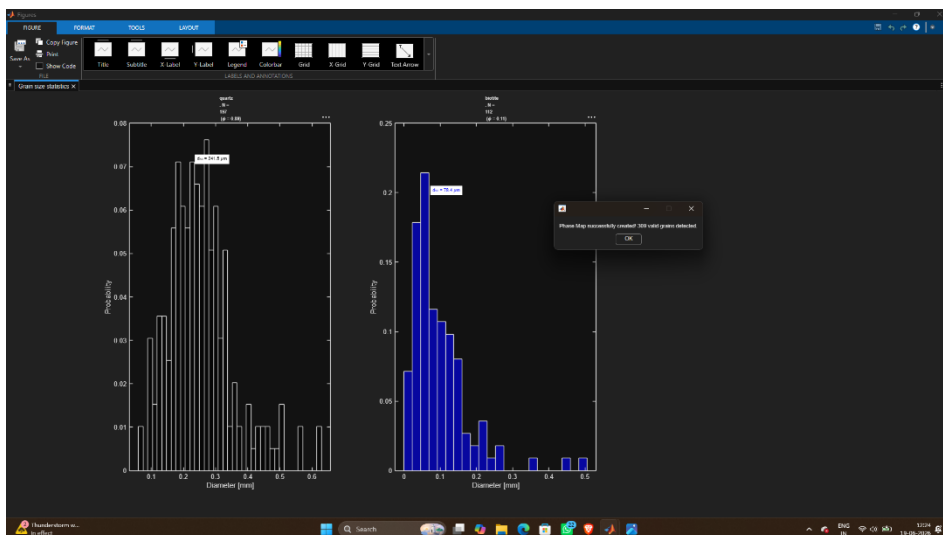
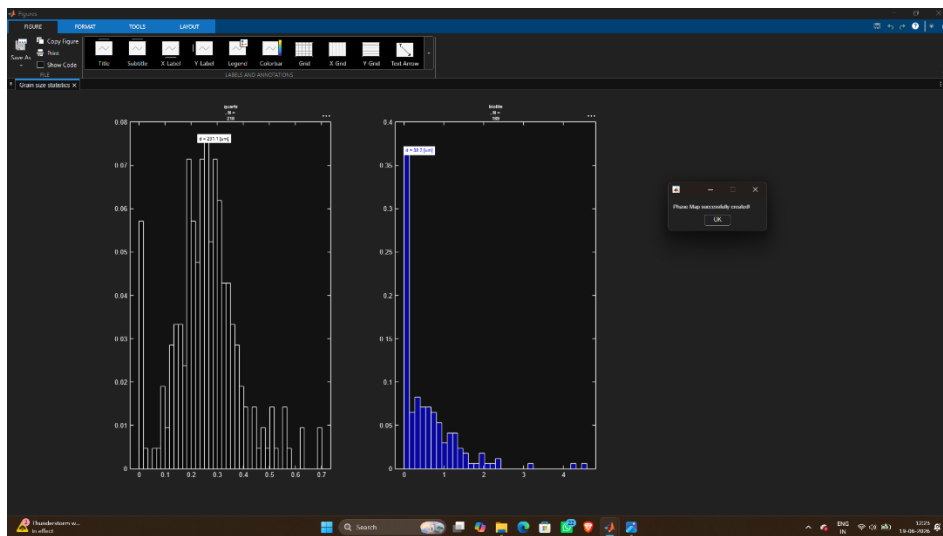
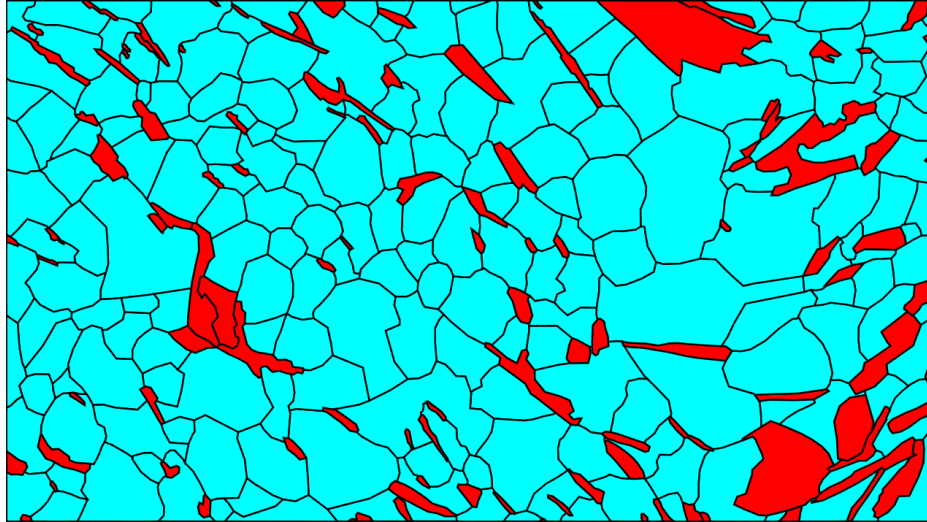
Appendix A: Additional Error Cases



Phasemap used in the Histogram Bin Edge Case



Output of the same phasemap after altering the code



Another testing Phasemap and the Output before and after Alteration of Codes